



Assignment 01

Multiple Minimum Support

Data Mining

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Introduction

This report is the documentation of the assignment on Multiple Minimum Support. It is an association mining rule, which is an upgrade from the standard Apriori because it also considers rules between rare item sets.

The dataset provided a listening history of music from [Last.fm](https://www.last.fm). It contains listening histories of 1,892 users and 17,632 artists. Each record also stores how many times each user listened to an artist.

Methodology

Preprocessing

The first step was to load user_artists.dat and artists.dat and merge them into one pandas dataframe. This contained artist, user, playcount, url and picture url.

	userID	artistID	weight	name	url	pictureURL
0	2	51	13883	Duran Duran	http://www.last.fm/music/Duran+Duran	http://userserve-ak.last.fm/serve/252/155668.jpg
1	2	52	11690	Morcheeba	http://www.last.fm/music/Morcheeba	http://userserve-ak.last.fm/serve/252/46005111...
2	2	53	11351	Air	http://www.last.fm/music/Air	http://userserve-ak.last.fm/serve/252/251119.jpg
3	2	54	10300	Hooverphonic	http://www.last.fm/music/Hooverphonic	http://userserve-ak.last.fm/serve/252/40553471...
4	2	55	8983	Kylie Minogue	http://www.last.fm/music/Kylie+Minogue	http://userserve-ak.last.fm/serve/252/12740835...

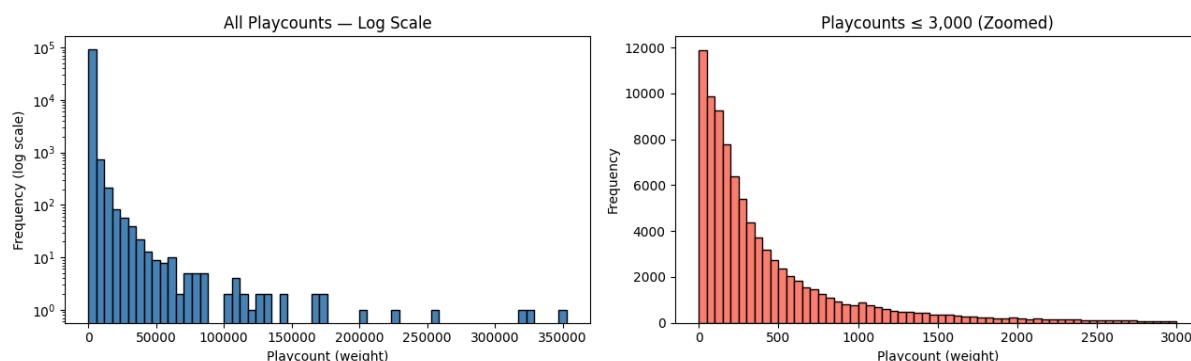
Transaction Definition

To extract transactions from the dataset, the set of artists with their playcounts were mapped on histograms to see the distribution. It was decided that the set of artists users listened to above a threshold would qualify as a transaction per user.

Threshold Justification

The threshold chosen based on the histogram was 200 as it was around the 45th percentile as shown in the figure below. This was just below the median of 260 playcounts. It removed one-off listening, removed noise and kept repeated listening.

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The table below shows that the average transaction length was 30.78 and the median, which is not influenced by outliers, was 32.

```
Threshold      : 200 plays
Total users    : 1755
Unique artists : 10791
Avg txn length : 30.78
Median txn length : 32.00
```

Top 15 Items by transaction

The table that follows shows the top 15 items by transaction frequency.

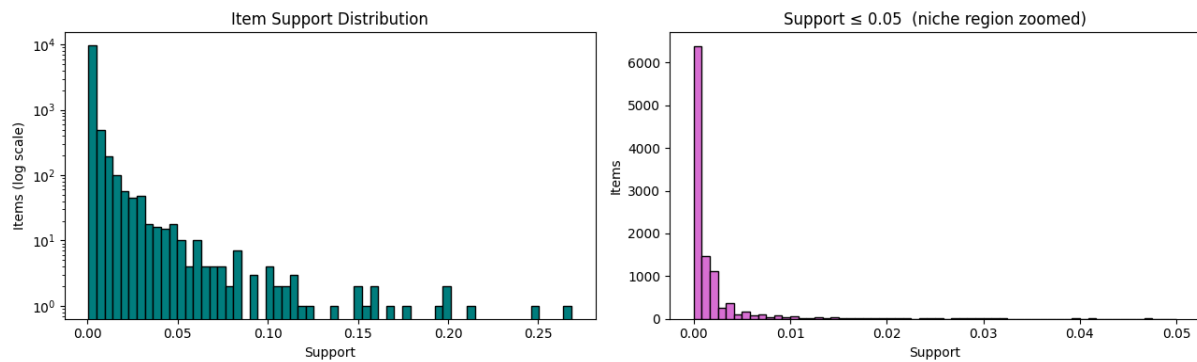
```
Top 15 items by transaction frequency:
Rank    Artist      Freq  Support
1       Lady Gaga   471   0.2684
2       Britney Spears 439   0.2501
3       Rihanna    370   0.2108
4       Katy Perry   353   0.2011
5       The Beatles  353   0.2011
6       Christina Aguilera 338   0.1926
7       Madonna     307   0.1749
8       Avril Lavigne 292   0.1664
9       Paramore    281   0.1601
10      Radiohead    280   0.1595
11      Beyoncé     268   0.1527
12      Muse         266   0.1516
13      Ke$ha        262   0.1493
14      Coldplay     242   0.1379
15      Kylie Minogue 217   0.1236
```

MIS Design

The figures below show the item support distribution and the support of items that are niche.

Based on the figures the Items with support ≥ 0.10 are classified as popular and they are 24. However 9717 Items with support < 0.005 are niche and rare.

So, support ≈ 0.02 separates popular mainstream artists from niche ones.



MIS Strategy

Based on the work of Liu et al. (1999) the method to calculate MIS is: $MIS(item) = \alpha \times \text{support}(item)$. Alpha is taken as 0.5 to include both niche and popular items. To keep F1 around a 100 items the MIS floor is 0.04. Based on this the MIS is calculated with the proportional strategy mentioned in the reference paper.

Phi Analysis

For phi analysis the support difference constraint, phi, is run at 0.001, 0.005, 0.010, 0.020, 0.050, 0.1, 0.2. The table below shows the analysis results.

phi	Candidates	Pruned	Remaining	Prune%
0.001	5,886	1,269	4,617	21.6%
0.005	5,886	979	4,907	16.6%
0.010	5,886	650	5,236	11.0%
0.020	5,886	261	5,625	4.4%
0.050	5,886	0	5,886	0.0%
0.100	5,886	0	5,886	0.0%
0.200	5,886	0	5,886	0.0%



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Based on the analysis, $\phi = 0.1$ is picked so that neither too many items are pruned and rare items are kept.

Multiple Minimum Support Algorithm

Seed List

The first step to running MMS Algorithm is to make the seed list. This is done with $MIS = 0.04$. Out of total items after the threshold which were 10791, 109 items are left L.

F1,F2,F3

The itemsets for F1, F2 and F3 are generated based on MMS rules. F1 has 109 itemsets, F2 has 194 and finally F3 has 341 itemsets. This means that items that were infrequent alone were frequent with other items. The below shows the details.

Level	Candidates	Frequent	Retained%
1	10,791	109	1.0%
2	5,236	194	3.7%
3	603	341	56.6%

Rules

The top 10 rules generated are shown below.

=== Top 10 Rules by Lift ===					
Antecedent	Consequent	Support	Confidence	Lift	
Leona Lewis	Rihanna, Mariah Carey	0.0405	0.6893	8.6411	
Rihanna, Leona Lewis	Mariah Carey	0.0405	0.7474	7.9493	
Lady Gaga, Mariah Carey	The Pussycat Dolls	0.0427	0.6356	7.6929	
Leona Lewis	Mariah Carey	0.0410	0.6990	7.4351	
Katy Perry, Ashley Tisdale	Miley Cyrus	0.0473	0.8557	7.4342	
Ashley Tisdale	Katy Perry, Miley Cyrus	0.0473	0.6694	7.3882	
Ke\$ha, Ashley Tisdale	Miley Cyrus	0.0416	0.8488	7.3748	
Mariah Carey, Beyoncé	The Pussycat Dolls	0.0416	0.6083	7.3629	
Selena Gomez & the Scene	Miley Cyrus	0.0444	0.8211	7.1334	
Lady Gaga, Ashley Tisdale	Miley Cyrus	0.0427	0.7895	6.8590	



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Conclusion

Interpretation

Users can be divided into different categories. One category is The Pop Girly who listens to all the mainstream hit music she sees on the billboard top 100. She's the girl who rocks to Katty Perry, Ashley Tisdale, Ke\$ha, Rihanna and Leona Lewis. What she needs is a 2000s Hot Girl Playlist recommended to her so she can have the shower concert of her dreams. An example of this is Leona Lewis ~> Rihanna, Mariah Carey.

Another category is of the Snooby Alt Music fan who only listens to tasteful music by British singers. This means Radiohead, Muse and Coldplay. They want to listen to something that moves their soul, so a playlist mix titled Fighting my inner demons - Brit Alt Rock is perfect for them. An example rule of this is Radiohead ~> Muse

Last is the category of Girlpower Teen Pop Music. These are teenagers in their cool girl era who are obsessed with Disney stars Miley Cyrus and Selena Gomez back when they were all the rage. What they need is a girl power playlist and recommendations to baddies like Taylor Swift and Camila Cabello. An example of this is Selena Gomez & the Scene ~> Miley Cyrus

Design Challenge

Rules were found by firstly making rules based on support, confidence and lift. The top 15 rules were scored by the $0.25 \times \text{support} + 0.35 \times \text{confidence} + 0.4 \times \text{lift_norm}$.

Redundant rules were eliminated. This means that if a F2 itemset already had the level of confidence then the F3 itemset with an extra item was removed from the rules, as it didn't add anything to the rule. After checking for redundancy 138 rules were eliminated.

Finally, the best 50 rules were found by finding the best rules with the formula above and setting the limit to 50. This whole analysis provided a way to check association rules between different artists to see similar taste patterns in users, to recommend them better aligned music.